Module 4 — the results of all exercises

Bioinformatik: metoder og anvendelse · for students who could not use ERDA

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This document shows the **real output of every exercise in the Module 4 workbook** that needs ERDA: everything kallisto and R print, and every figure. It was produced by running the workbook code exactly as written, on the same data. Read it next to the workbook — the section numbers (§) are the workbook's: https://mutwil.github.io/bioinformatik-module4-transcriptomics/

The files needed for Cytoscape (Sessions 5–6) can be downloaded as well:

- Session 5 network (CESA7): https://mutwil.github.io/bioinformatik-module4-transcriptomics/data/exercise_outputs/for_cytoscape_AT5G17420.tsv
- Mini-project, three-layer network of each gene: https://mutwil.github.io/bioinformatik-module4-transcriptomics/data/exercise_outputs/network_AT5G17420.tsv (replace `AT5G17420` by the AGI code of any of the ten genes in §6.1)

Session 1 — kallisto (§1.4–1.9)

This part shows the **real output** of the Session 1 kallisto commands (the same data, commands and kallisto version, 0.46.1), and explains every part of it. You can also download the result files and open them in Excel or R:

- leaf (SRR3581681): abundance.tsv · run_info.json
- internode (SRR3581705): abundance.tsv · run_info.json
- the problem sample (SRR34314723): abundance.tsv · run_info.json

Step 1 — `kallisto index`: building the index

```
[build] loading fasta file TAIR10_cds.fa
[build] k-mer length: 31
[build] warning: replaced 8 non-ACGUT characters in the input sequence
        with pseudorandom nucleotides
[build] counting k-mers ... done.
[build] building target de Bruijn graph ...  done
[build] creating equivalence classes ...  done
[build] target de Bruijn graph has 86480 contigs and contains 31977016 k-mers
```

- kallisto reads the 27,416 coding sequences and cuts them into **31-letter words (*k*-mers)** — about 32 million different ones.
- It stores them in a **de Bruijn graph**: a compact map of which *k*-mers follow each other and **which transcripts contain each *k*-mer**. This is the lookup table used in the next step.
- The warning is harmless: 8 letters in the FASTA were not A, C, G or T (e.g. `N`) and were replaced at random.

Step 2 — `kallisto quant`: what it prints

Leaf sample (2 million reads):

```
[quant] fragment length distribution is truncated gaussian with mean = 200, sd = 20
[index] k-mer length: 31
[index] number of targets: 27,416
[index] number of k-mers: 31,977,016
[index] number of equivalence classes: 35,220
[quant] running in single-end mode
[quant] will process file 1: reads/SRR3581681.fastq.gz
[quant] finding pseudoalignments for the reads ... done
[quant] processed 2,000,000 reads, 1,362,175 reads pseudoaligned
[   em] quantifying the abundances ... done
[   em] the Expectation-Maximization algorithm ran for 525 rounds
```

Line	Meaning
`fragment length … mean = 200, sd = 20`	the values we gave with `-l 200 -s 20`; for single-end reads kallisto cannot measure the fragment length itself
`number of targets: 27,416`	the transcripts in the index — what reads can be assigned to
`equivalence classes`	groups of transcripts that share the same *k*-mers; a read that fits several transcripts is assigned to such a

Line	Meaning
	group
`processed 2,000,000 reads, 1,362,175 reads pseudoaligned`	**the key quality line**: 68 % of the reads matched the transcriptome
`Expectation-Maximization … 525 rounds`	reads that fit several transcripts are shared out between them by an iterative algorithm (EM) until the estimates stop changing

The problem sample prints `processed 30 reads, 6 reads pseudoaligned` — only 30 reads, and only 20 % of them match.

Step 3 — the three output files

Each sample gets its own folder (e.g. `results/SRR3581681/`) with:

File	What it contains
`abundance.tsv`	the results: one row per transcript (27,416 rows) — **this is the file you use**
`abundance.h5`	the same results in a binary format (HDF5), for other programs such as *sleuth*
`run_info.json`	a short report on the run: settings, number of reads, versions

`run_info.json`

Field	Leaf (SRR3581681)	Problem sample (SRR34314723)	Meaning
`n_targets`	27416	27416	transcripts in the index
`n_bootstraps`	0	0	bootstrap repeats (not used here)
`n_processed`	2000000	**30**	reads read from the FASTQ file
`n_pseudoaligned`	1362175	**6**	reads that matched at least one transcript
`n_unique`	1335593	6	reads that matched exactly **one** transcript
`p_pseudoaligned`	68.1	**20.0**	% of reads that matched
`p_unique`	66.8	20.0	% that matched exactly one transcript
`kallisto_version` / `index_version`	0.46.1 / 10	0.46.1 / 10	software versions (reproducibility)

Field	Leaf (SRR3581681)	Problem sample (SRR34314723)	Meaning
`start_time` / `call`			when the run started, and the exact command

How to read it: a good sample has **millions of reads** and typically **60–90 % pseudoaligned**. The problem sample has 30 reads, of which 20 % match: the sequencing failed or the file is truncated — and, as the SRA shows, it is a test run from rice and sorghum, so the wrong organism for an Arabidopsis index.

`abundance.tsv`

The first rows of the leaf sample:

target_id	length	eff_length	est_counts	tpm
AT1G50920.1	2016	1817	152	35.5993
AT1G36960.1	546	347	0	0
AT1G44020.1	1734	1535	0	0
AT1G15970.1	1059	860	12	5.93794
AT1G73440.1	765	566	17.0106	12.7896

Column	Meaning
`target_id`	the transcript (here: AGI code + `.1` = first gene model)
`length`	length of the transcript in nucleotides
`eff_length`	**effective length**: the number of positions where a 200-nt fragment can start, ≈ length − 200 + 1 (2016 → 1817)
`est_counts`	**estimated number of reads** from this transcript. Can be a decimal number (17.0106) because a read that fits several transcripts is shared between them
`tpm`	**transcripts per million**: the counts corrected for transcript length and sequencing depth. All TPM values of a sample add up to 1,000,000

TPM is calculated in two steps — the same as in lecture M4F2:

1. divide each transcript's `est_counts` by its `eff_length` (corrects for length: long transcripts produce more fragments);
2. divide by the sum of these values over all transcripts and multiply by 1,000,000 (corrects for sequencing depth).

For example, PsaK in the leaf: 483 reads / 194 nt, scaled so that the sample sums to one million, gives **1059.49 TPM**.

Three genes in the two organs:

Gene	Transcript	Leaf: est_counts	Leaf: TPM	Internode: est_counts	Internode: TPM
PsaK (photosynthesis)	AT1G30380.1	483	1059.5	459	1144.7
CESA4 (secondary wall)	AT5G44030.1	14	2.0	892	146.2
CESA7 (secondary wall)	AT5G17420.1	4	0.6	1045	175.4

- PsaK has a **short** transcript (393 nt), so 483 reads already give a high TPM.
- CESA4 and CESA7 are about **70–300× higher in the internode** — the stem, full of secondary cell walls — than in the leaf, which is what you analyse in §1.10.
- In the leaf, CESA4 has only 14 reads: lowly expressed genes get few reads, so their TPM is less precise.

Session 1 — Is CESA4 expressed where you would expect? (§1.10)

TPM of CESA4 in the two samples (first 2 million reads of each):

	TPM
Leaf	2.01888
Internode	146.23800

CESA4 is about 70× higher in the stem internode than in the leaf — it makes cellulose for the secondary cell walls of the stem.

Session 2 — EDA and preprocessing

2.1 The expression matrix

Which organ each sequencing run comes from (first rows of the sample annotation), the size of the matrix (27,619 genes × 56 samples) and its first rows and columns:

```
   Comment[ENA_RUN] Characteristics[organism part]
1        ERR3333388                           petal
3        ERR3333389                       stem node
5        ERR3333390                  stem internode
7        ERR3333391                    cauline leaf
9        ERR3333392                    rosette leaf
11       ERR3333393                    rosette leaf
[1] 27619    56
           adult vascular leaf rosette leaf rosette leaf rosette leaf
AT1G01010                11.0            4            6         13.0
AT1G01020                 0.0            0            0          0.8
AT1G01030                 0.4            0            6          5.0
AT1G01040                 6.0            8            6         11.0
AT1G01050                67.0           89           86        134.0
```

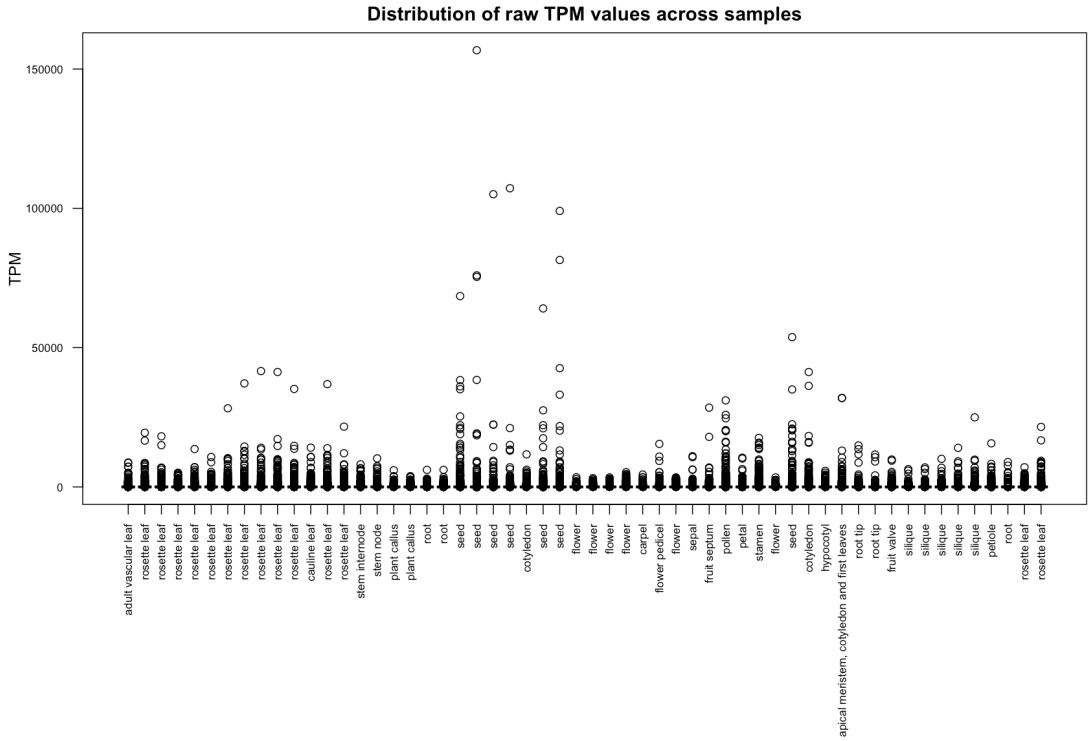
2.2 How many genes are expressed?

```
27619 genes are present in the original expression matrix.
25835 genes are expressed at TPM > 1 in at least one sample.
Thus, we removed 1784 lowly expressed genes.
```

With threshold 10: 21,305 genes; with threshold 100: 7,323 genes.

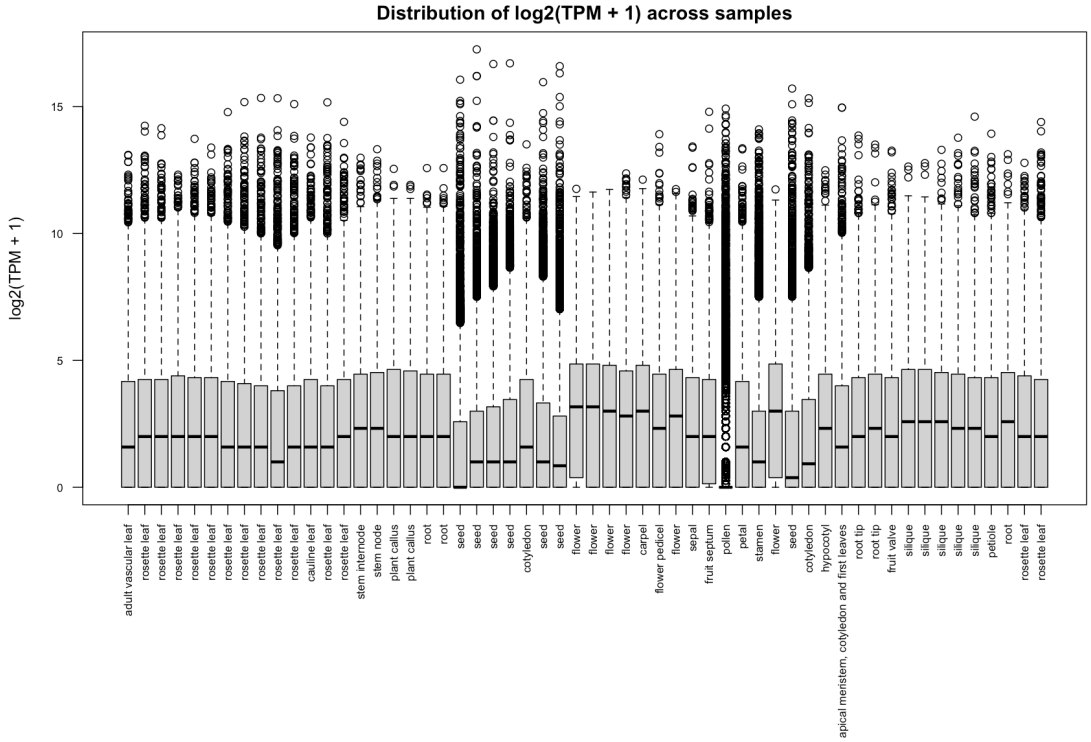
2.3 Distributions: raw TPM and log2(TPM + 1)

Raw TPM — a few genes with enormous values squash every box flat:



Raw TPM

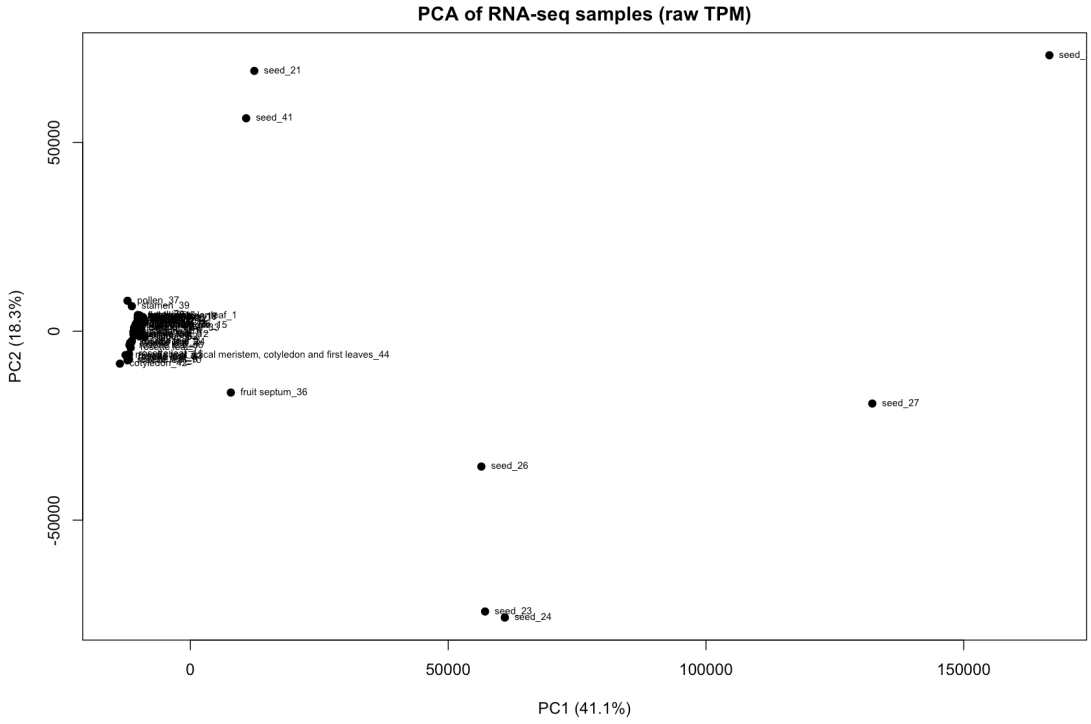
After log2(TPM + 1) — now the distributions can be compared. Note pollen (median 0) and seed (low median, very long upper tail):



log2(TPM+1)

2.4 PCA on raw TPM

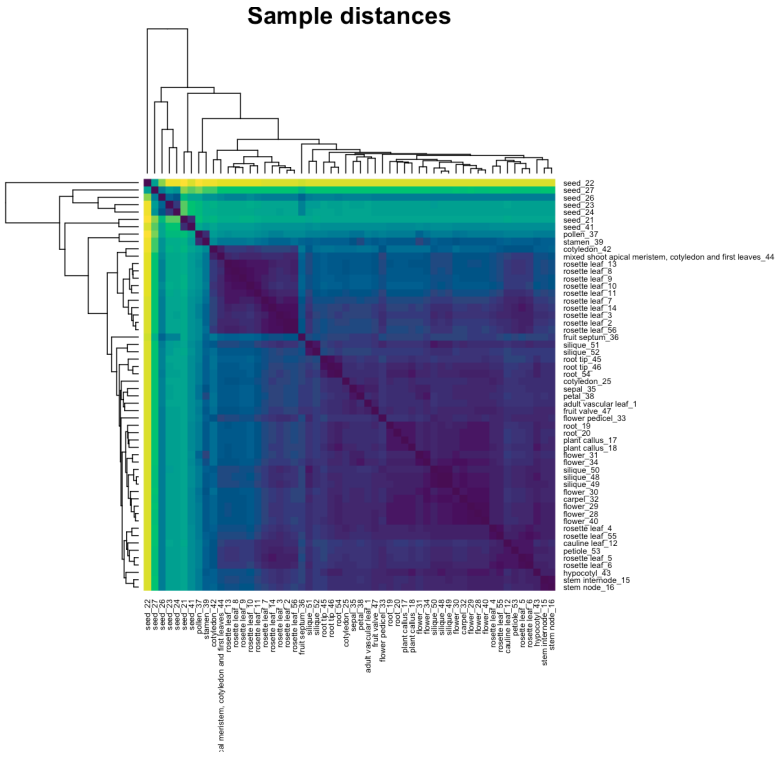
The seed samples dominate PC1; everything else is lumped together.



PCA raw

2.5 Sample distance heatmap (raw TPM, Euclidean distance)

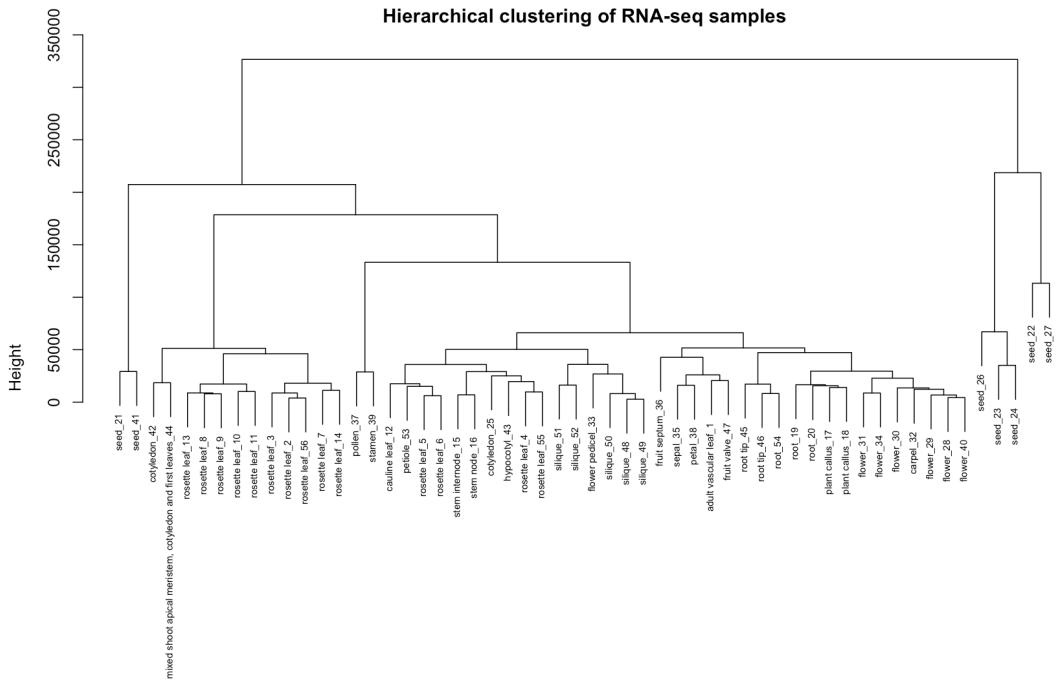
Dark = similar, bright = different. The seed samples form their own block.



Heatmap raw

2.6 Hierarchical clustering (raw TPM, Ward)

The first split separates five seed samples from all other samples.



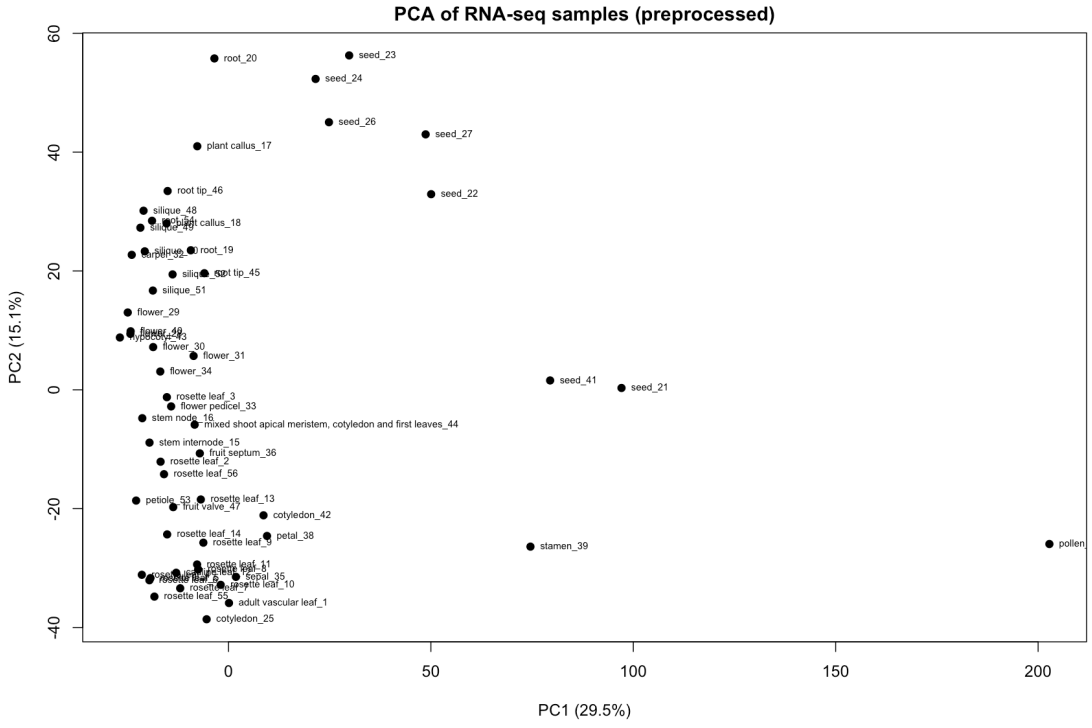
Dendrogram raw

2.7 After preprocessing

Genes kept after filtering and selecting the 5,000 most variable genes:

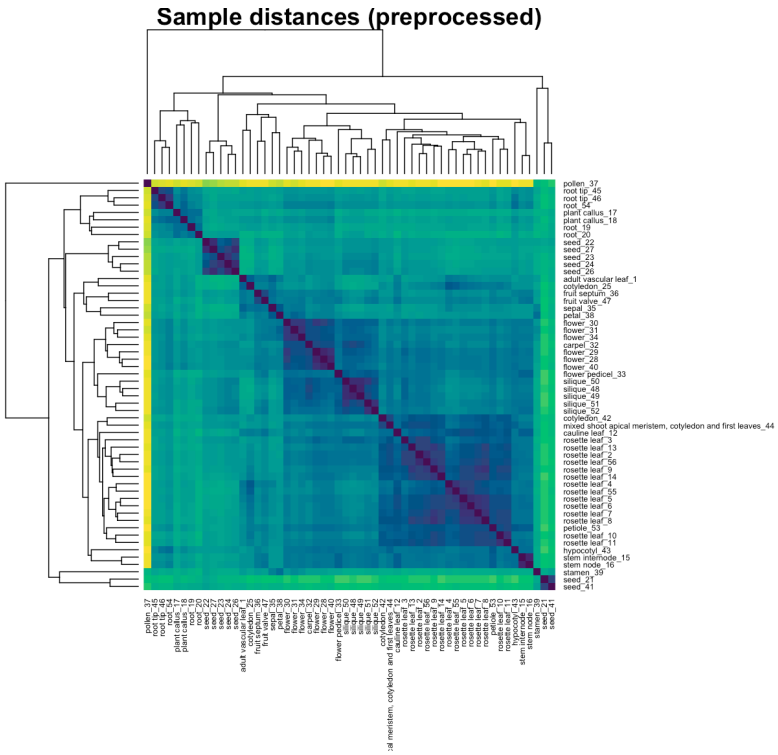
```
[1] 5000   56
```

PCA after preprocessing — pollen (and seed, stamen) stand apart on PC1; PC2 orders the other organs (roots and callus at the top, flowers and siliques in the middle, leaves at the bottom):



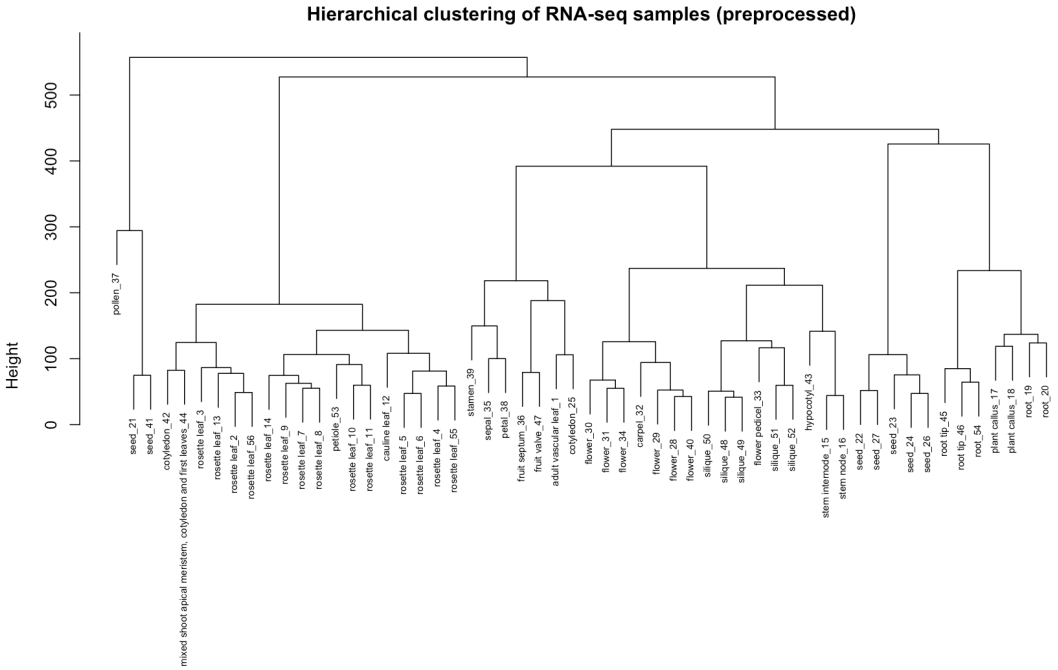
PCA preprocessed

Sample distance heatmap after preprocessing — the organ groups are much clearer:



Heatmap preprocessed

Dendrogram after preprocessing:



Dendrogram preprocessed

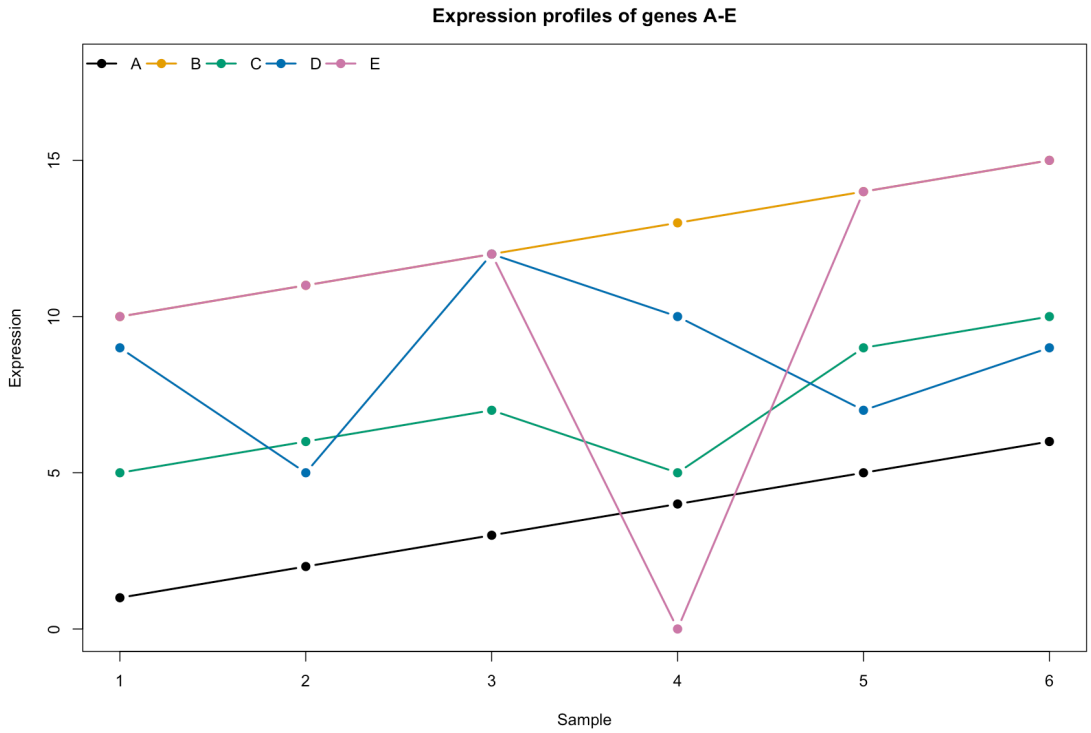
Session 3 — Co-expression

3.1 Five made-up genes

The PCC of gene A with each of the other genes:

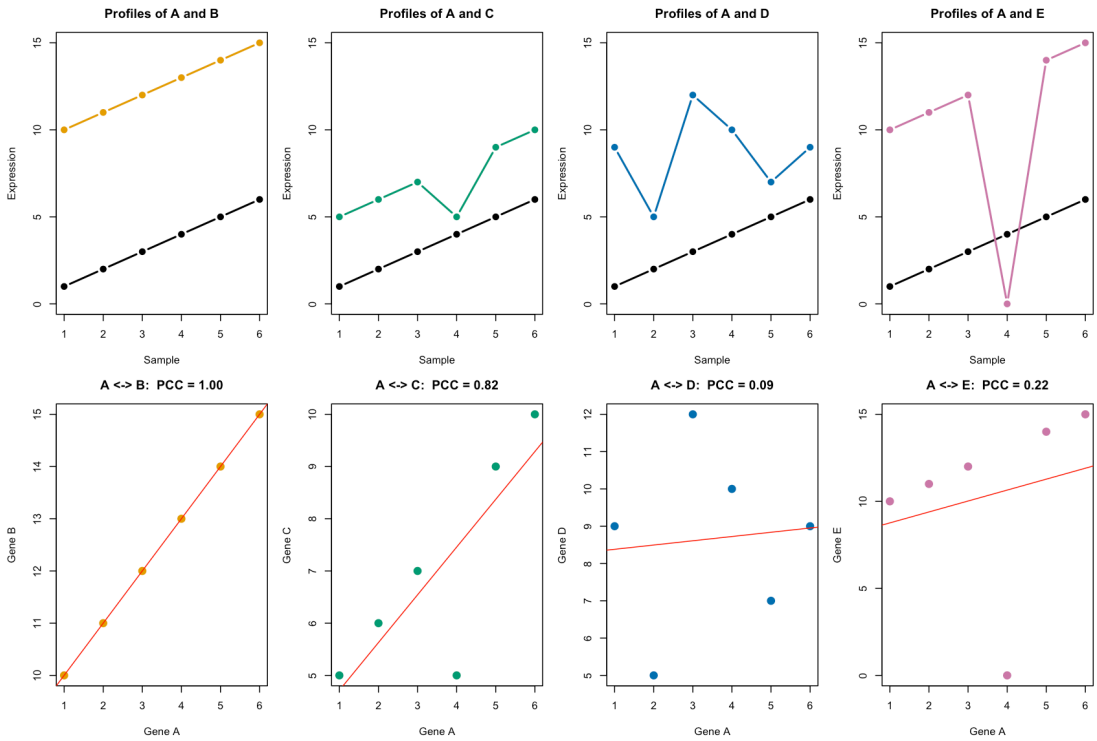
```
A <-> B:  r =  1.000   (p = 0.00e+00)
A <-> C:  r =  0.815   (p = 4.80e-02)
A <-> D:  r =  0.088   (p = 8.68e-01)
A <-> E:  r =  0.218   (p = 6.78e-01)
```

The five expression profiles:



Profiles

Profiles of A next to each gene (top) and A plotted against each gene with the regression line (bottom). E follows A in five of six samples, but one outlier pulls the PCC down to 0.22:



Pairwise

3.2 The annotated matrix

```
[1] 27619    23
       gene                                        description
1 AT1G01010                    NAC domain containing protein 1
2 AT1G01020                                  Arv1-like protein
3 AT1G01030 AP2/B3-like transcriptional factor family protein
4 AT1G01040                                    dicer-like 1
5 AT1G01050                                pyrophosphorylase 1
6 AT1G01060              Homeodomain-like superfamily protein
```

3.3 The genes co-expressed with PsaK (top 40 of 100)

```
gene      PCC   description
AT1G30380 1.000 photosystem I subunit K
AT4G21280 0.995 photosystem II subunit QA
AT1G15820 0.992 light harvesting complex photosystem II subunit 6
AT4G01150 0.990
AT4G28750 0.990 Photosystem I reaction centre subunit IV / PsaE pr
AT4G09650 0.989 ATP synthase delta-subunit gene
AT1G52230 0.988 photosystem I subunit H2
AT1G06680 0.985 photosystem II subunit P-1
AT1G31330 0.984 photosystem I subunit F
AT1G03130 0.983 photosystem I subunit D-2
AT1G08380 0.981 photosystem I subunit O
AT1G61520 0.981 photosystem I light harvesting complex gene 3
AT1G20340 0.980 Cupredoxin superfamily protein
AT3G46780 0.980 plastid transcriptionally active 16
AT3G47470 0.977 light-harvesting chlorophyll-protein complex I sub
AT2G06520 0.976 photosystem II subunit X
AT2G30570 0.975 photosystem II reaction center W
AT2G46820 0.975 photosystem I P subunit
AT1G67740 0.974 photosystem II BY
AT1G60950 0.973 2Fe-2S ferredoxin-like superfamily protein
```

```
 AT4G10340 0.973 light harvesting complex of photosystem II 5
 AT4G14890 0.970 2Fe-2S ferredoxin-like superfamily protein
 AT5G23060 0.970 calcium sensing receptor
 AT1G75350 0.969 Ribosomal protein L31
 AT2G39470 0.969 PsbP-like protein 2
 AT3G04790 0.967 Ribose 5-phosphate isomerase, type A protein
 AT5G13630 0.967 magnesium-chelatase subunit chlH, chloroplast, put
 AT5G01530 0.965 light harvesting complex photosystem II
 AT3G54890 0.964 photosystem I light harvesting complex gene 1
 AT1G76450 0.962 Photosystem II reaction center PsbP family protein
 AT1G75690 0.961 DnaJ/Hsp40 cysteine-rich domain superfamily protei
 AT2G05070 0.960 photosystem II light harvesting complex gene 2.2
 AT3G01480 0.960 cyclophilin 38
 AT3G56940 0.960 dicarboxylate diiron protein, putative (Crd1)
 AT4G02770 0.960 photosystem I subunit D-1
 AT1G05190 0.959 Ribosomal protein L6 family
 AT3G54210 0.958 Ribosomal protein L17 family protein
 AT2G43560 0.957 FKBP-like peptidyl-prolyl cis-trans isomerase fami
 AT4G01050 0.957 thylakoid rhodanese-like
 AT1G51400 0.956 Photosystem II 5 kD protein
 AT1G55670 0.956 photosystem I subunit G
… (59 more lines)
```

Session 5 — The CESA7 network

5.2 The genes co-expressed with CESA7 (top 40 of 100)

gene	PCC	description
AT5G17420_CESA7	1.000	Cellulose synthase family protein
AT4G18780_CESA8	0.962	cellulose synthase family protein
AT2G38080_IRX12	0.958	Laccase/Diphenol oxidase family protein
AT2G40320_TBL33	0.955	TRICHOME BIREFRINGENCE-LIKE 33
AT5G44030_CESA4	0.952	cellulose synthase A4
AT3G50220_IRX15	0.951	Protein of unknown function (DUF579)
AT1G27440_IRX10	0.940	Exostosin family protein
AT3G16920_CTL2	0.940	chitinase-like protein 2
AT5G60720_nan	0.937	Protein of unknown function, DUF547
AT3G55990_ESK1	0.934	Plant protein of unknown function (DUF828)
AT4G28500_NAC073	0.934	NAC domain containing protein 73
AT5G60020_LAC17	0.928	laccase 17
AT5G54690_GAUT12	0.920	galacturonosyltransferase 12
AT3G18660_GUX1	0.918	plant glycogenin-like starch initiation protein 1
AT3G62020_GLP10	0.918	germin-like protein 10
AT1G09610_GXM1	0.917	Protein of unknown function (DUF579)
AT2G28315_UXT1	0.916	Nucleotide/sugar transporter family protein
AT2G29130_LAC2	0.916	laccase 2
AT5G12870_MYB46	0.916	myb domain protein 46
AT1G08340_ROPGAP5	0.911	Rho GTPase activating protein with PAK-box/P21-Rho
AT5G03170_FLA11	0.910	FASCICLIN-like arabinogalactan-protein 11
AT1G63910_AtMYB103	0.907	myb domain protein 103
AT3G15050_IQD10	0.905	IQ-domain 10
AT5G60490_FLA12	0.904	FASCICLIN-like arabinogalactan-protein 12
AT5G01360_TBL3	0.897	Plant protein of unknown function (DUF828)
AT3G22415_nan	0.886	
AT5G16600_MYB43	0.884	myb domain protein 43
AT5G40020_nan	0.884	Pathogenesis-related thaumatin superfamily protein
AT1G28470_NAC010	0.881	NAC domain containing protein 10
AT1G32100_PRR1	0.878	pinoresinol reductase 1
AT5G37478_nan	0.876	TPX2 (targeting protein for Xklp2) protein family
AT2G31930_nan	0.871	
AT1G22480_nan	0.869	Cupredoxin superfamily protein
AT2G46770_NAC043	0.866	NAC (No Apical Meristem) domain transcriptional re
AT5G45970_ARAC2	0.862	RAC-like 2
AT4G17220_MAP70.5	0.855	microtubule-associated proteins 70-5
AT2G03200_nan	0.849	Eukaryotic aspartyl protease family protein
AT5G02090_nan	0.849	
AT5G14510_nan	0.849	ARM repeat superfamily protein
AT4G22680_MYB85	0.847	myb domain protein 85
AT5G05390_LAC12	0.847	laccase 12

… (59 more lines)

5.3 The network

```
Saved 195 edges (PCC > 0.9) among 100 genes to for_cytoscape_AT5G17420.tsv
```

The 195 edges are in `for_cytoscape_AT5G17420.tsv` (download link at the top).

Mini-project — example: CESA7 (§6.4–6.7)

The same steps give a different result for each of the ten genes; the network files of all ten are available for download (link at the top).

Q4: CESA7's first neighbours

```
195 co-expression edges among 57 genes
 gene               PCC   description
 AT4G18780_CESA8    0.962 cellulose synthase family protein
 AT2G38080_IRX12    0.958 Laccase/Diphenol oxidase family protein
 AT2G40320_TBL33    0.955 TRICHOME BIREFRINGENCE-LIKE 33
 AT5G44030_CESA4    0.952 cellulose synthase A4
 AT3G50220_IRX15    0.951 Protein of unknown function (DUF579)
 AT1G27440_IRX10    0.940 Exostosin family protein
 AT3G16920_CTL2     0.940 chitinase-like protein 2
 AT5G60720_nan      0.937 Protein of unknown function, DUF547
 AT4G28500_NAC073   0.934 NAC domain containing protein 73
 AT3G55990_ESK1     0.934 Plant protein of unknown function (DUF828)
 AT5G60020_LAC17    0.928 laccase 17
 AT5G54690_GAUT12   0.920 galacturonosyltransferase 12
 AT3G62020_GLP10    0.918 germin-like protein 10
 AT3G18660_GUX1     0.918 plant glycogenin-like starch initiation protein 1
 AT1G09610_GXM1     0.917 Protein of unknown function (DUF579)
 AT5G12870_MYB46    0.916 myb domain protein 46
 AT2G28315_UXT1     0.916 Nucleotide/sugar transporter family protein
 AT2G29130_LAC2     0.916 laccase 2
 AT1G08340_ROPGAP5  0.911 Rho GTPase activating protein with PAK-box/P21-Rho
 AT5G03170_FLA11    0.910 FASCICLIN-like arabinogalactan-protein 11
 AT1G63910_AtMYB103 0.907 myb domain protein 103
 AT3G15050_IQD10    0.905 IQ-domain 10
 AT5G60490_FLA12    0.904 FASCICLIN-like arabinogalactan-protein 12
```

Q5: CESA7's interaction partners (BioGRID)

```
   partner    symbol               experiment co_expressed
 AT5G44030     CESA4 Affinity Capture-Western         TRUE
 AT4G18780      IRX1 Affinity Capture-Western         TRUE
 AT1G01510        AN               Two-hybrid        FALSE
 AT4G32410     CESA1               Two-hybrid        FALSE
 AT5G05170      CEV1               Two-hybrid        FALSE
 AT5G64740     CESA6               Two-hybrid        FALSE
 AT5G44030     CESA4               Two-hybrid         TRUE
 AT4G18780      IRX1               Two-hybrid         TRUE
 AT4G32410     CESA1                      PCA        FALSE
 AT5G05170      CEV1                      PCA        FALSE
 AT5G64740     CESA6                      PCA        FALSE
 AT1G43790      TED6 Affinity Capture-Western         TRUE
 AT1G43790      TED6          Co-purification         TRUE
 AT1G43790      TED6                      PCA         TRUE
 AT5G06320      NHL3                      PCA        FALSE
 AT3G12180 AT3G12180                      PCA        FALSE
 AT1G34640 AT1G34640                      PCA        FALSE
 AT5G49540 AT5G49540                      PCA        FALSE
 AT1G32740 AT1G32740                      PCA        FALSE
 AT5G44030     CESA4                      PCA         TRUE
 AT4G18780      IRX1                      PCA         TRUE
31 protein-protein interactions in the PPI layer
```

Q6: transcription factors that regulate CESA7 and its neighbours (first 15 lines)

TF	regulates_yours	n_genes	genes	effect	evidence	reference
MYB46	TRUE	3	CESA4, CESA7, CESA8	Activation	targeted study	Kim et al.
ZAT7	TRUE	1	CESA7	Activation		
ILR3	TRUE	7	AT5G60720, CESA4, CESA7, CESA8, IRX10, L	unknown		
AT1G61730	TRUE	6	CESA4, CESA7, GUX1, GXM3, IRX10, MYB46	unknown		
AT2G44730	TRUE	6	AT5G60720, CESA4, CESA7, IRX10, LAC17, M	unknown		
ATE2F2	TRUE	6	AT2G38080, AT5G60720, CESA4, CESA7, LAC1	unknown		
MYB73	TRUE	6	CESA7, CESA8, GUX1, IRX10, LAC17, MYB46	unknown		
ARF9	TRUE	4	CESA4, CESA7, CESA8, MYB46	unknown		
BZIP30	TRUE	3	CESA4, CESA7, LAC17	unknown		
ERF9	TRUE	3	CESA4, CESA7, GXM3	unknown		
PLT3	TRUE	3	CESA4, CESA7, MYB46	unknown		
DREB2	TRUE	2	CESA7, MYB46	unknown		
DREB26	TRUE	2	CESA7, GUX1	unknown		
FLO2	TRUE	1	CESA7	unknown		
BES1	FALSE	2	CESA4, CESA8	Activation		

… (15 more lines)

The three layers of the network

